Date : Name of the Experiment : BCD-7 - Segment display

Date : 9/7/24 Name of the Experiment : ADDERS Page No. : 5

Aim : To design and verify the output of a half adder and full adder circuit and by using them in adder design a four bit parallel adder.

Components :

1. Bread board
2. Power Supply
3. Integrated Circuits [7408, 7432, 7486]
4. Connecting wires
5. Multimeter

Theory : The most basic arthematic Operation used in any circuit is binary addition. The four elementary operations are

$0 + 0 = 0$

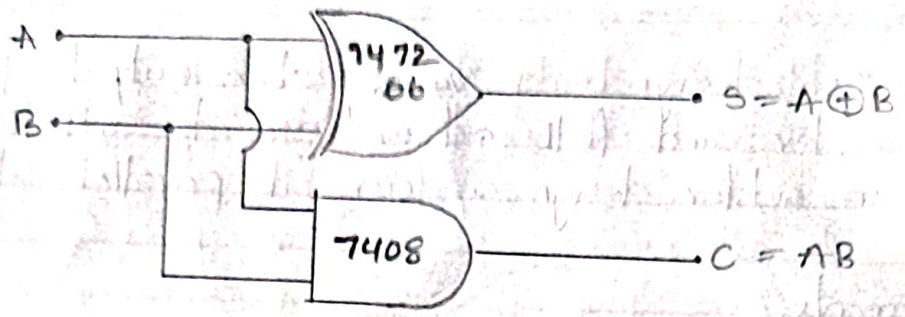
$0 + 1 = 1$

$1 + 0 = 1$

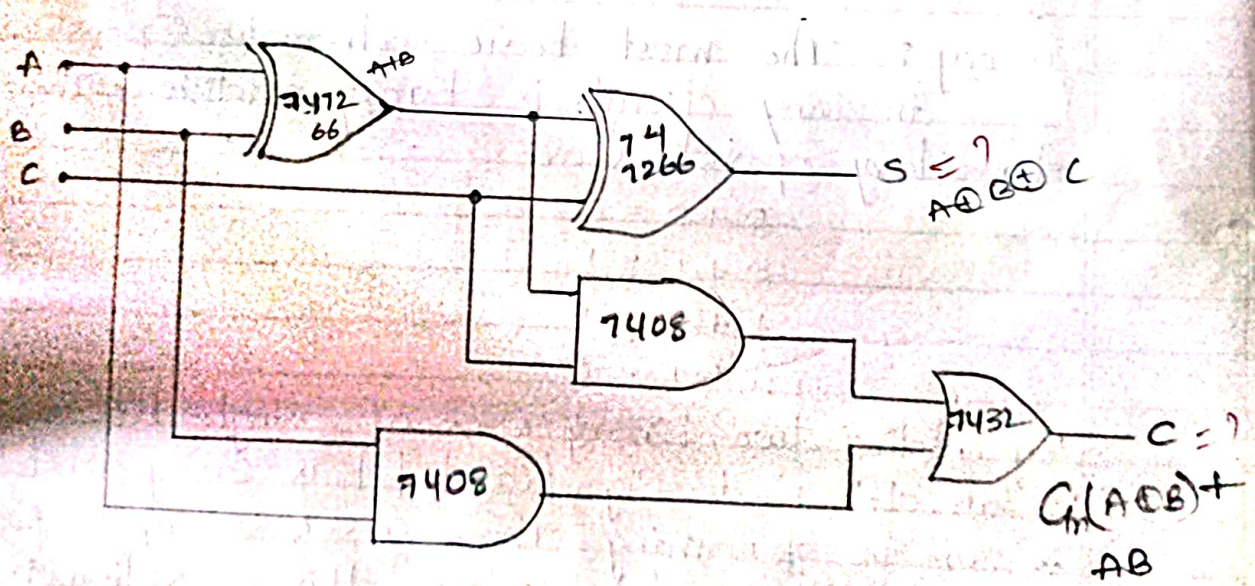
$1 + 1 = 0$

By adding two binary ones of single bit we get 2 bits the most significant bit is carry and LSB is sum in remaining three cases carry is zero. The circuit which perform this operation is called half adder where as in full adder we add 3 bits simultaneously.

Half ADDER



FULL ADDER



Half adder: Half adders needs two inputs arranged and truth table corresponding to this is:

x	y	x+y	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

By observing the truth table we can say that sum is implement using X-OR carry implement using AND gate

Full adder:

It requires three input A, B of cin [Carry from previous] and result in two out sum of carry truth table corresponding to this is:

A	B	Cin	S	cin+1
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Half adder

Input		output			
A	B	sum	practical	carry	practical
0	0	0	0.41	0	0.31
0	1	1	4.31	0	0.5
1	0	1	4.02	0	0.31
1	1	0	0.62	1	4.83

Input			output			
A	B	C	sum	practical	carry	practical
0	0	0	0	0.01	0	0.1
0	0	0	1	4.8	0	0.02
0	1	0	1	4.7	0	0.01
0	1	0	0	0.01	1	4.8
1	0	1	1	4.5	0	0.00
1	0	1	0	0.05	1	4.8
1	1	1	0	0.04	1	4.6
1	1	1	1	4.9	1	4.67

By observing sums, where ever odd numbers of ones one input we get output as high so we can implement sum using X-OR gate.

$$F = A \oplus B \oplus C_{in}$$

Carry $C_{in+1} = \sum m(3,5,6,7)$

$$= \bar{A}B[C_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in}]$$

$$= AC_{in}[\bar{A}B + A\bar{B}] + AB[\bar{C}_{in} + C_{in}]$$

$$= C_{in}[A \oplus B] + AB$$

We can add two four bit binary inputs by connecting four full adders one other the other.

Procedure:

1. Connect the circuit on bread board.
2. Give a 5V dc supply to circuit (IC's).
3. Now gives the input to the circuit [inputs of different combinations] and check the output.
4. Compare the obtained results with available truth table.

Precautions:

1. Connections are given as per the logic diagram.
2. Logical inputs are given as per the logic diagram.
3. Observe the outputs and verify the truth table.

Result:

Thus the Half adder, full adder, half circuits are designed and their truth tables are verified.

6/7/24